

Pseudo Likelihood

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In this document, the pseudo Likelihood estimation (PLE) for the Ising and Potts models used in the `spininfer` library is defined and derived. First, the gradient derivations are presented for the Ising model and subsequently generalized to the Potts model.

1 Ising Model

This section provides a derivation of the gradients used in the pseudo likelihood estimation (PLE) for the Ising model. The relevant known equations are:

$$H(\boldsymbol{\sigma}) = - \sum_i h_i \sigma_i - \sum_i \sum_{j < i} J_{ij} \sigma_i \sigma_j, \quad (1)$$

$$p(\boldsymbol{\sigma}) = \frac{1}{Z} e^{-H(\boldsymbol{\sigma})} = \frac{1}{Z} \exp \left(\sum_i h_i \sigma_i + \sum_i \sum_{j < i} J_{ij} \sigma_i \sigma_j \right), \quad (2)$$

$$Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})}. \quad (3)$$

Assuming symmetric couplings without self-interactions ($J_{ij} = J_{ji}$ and $J_{ii} = 0$), the interaction sum can be rewritten symmetrically. The Hamiltonian can then be split into the energy contributions of a single site σ_i and the remainder of the system:

$$H(\boldsymbol{\sigma}) = H(\boldsymbol{\sigma} \setminus \sigma_i) + H(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) = H(\boldsymbol{\sigma} \setminus \sigma_i) - h_i \sigma_i - \sigma_i \sum_{j \neq i} J_{ij} \sigma_j. \quad (4)$$

Using the definition of conditional probability, we write the conditional distribution of site i given the state of all other sites as:

$$P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) = \frac{P(\sigma_i, \boldsymbol{\sigma} \setminus \sigma_i)}{P(\boldsymbol{\sigma} \setminus \sigma_i)} = \frac{P(\sigma_i, \boldsymbol{\sigma} \setminus \sigma_i)}{\sum_{\sigma'_i \in \{-1, +1\}} P(\sigma'_i, \boldsymbol{\sigma} \setminus \sigma_i)}. \quad (5)$$

Defining the effective field acting on site i as:

$$\theta_i = h_i + \sum_{j \neq i} J_{ij} \sigma_j, \quad (6)$$

and inserting the Boltzmann distribution into Equation 5, we get:

$$\begin{aligned} P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) &= \frac{e^{\sigma_i \theta_i}}{e^{\theta_i} + e^{-\theta_i}} \\ &= \frac{e^{\sigma_i \theta_i}}{e^{\sigma_i \theta_i} + e^{-\sigma_i \theta_i}} = \frac{1}{1 + e^{-2\sigma_i \theta_i}}. \end{aligned} \quad (7)$$

Now, in the pseudo likelihood method, we aim to optimize the log conditional probabilities. The derivative of the log conditional probability with respect to the field h_a is given by:

$$\begin{aligned} \frac{\partial \ln P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i)}{\partial h_a} &= \frac{\partial -\ln(1 + e^{-2\sigma_i \theta_i})}{\partial h_a} \\ &= \delta_{ia} \frac{2\sigma_i e^{-2\sigma_i \theta_i}}{1 + e^{-2\sigma_i \theta_i}} = \delta_{ia} \frac{2\sigma_i}{1 + e^{2\sigma_i \theta_i}} = \delta_{ia} \sigma_i (1 - \tanh \sigma_i \theta_i), \end{aligned} \quad (8)$$

where in the last equality the identity $\frac{2e^{-2x}}{1+e^{-2x}} = 1 - \tanh x$ was used. Using the property $\tanh(-\sigma_i \theta_i) = -\sigma_i \tanh(\theta_i)$ and $\sigma_i^2 = 1$, the derivative simplifies to:

$$\frac{\partial \ln P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i)}{\partial h_a} = \delta_{ia} \sigma_i (1 - \sigma_i \tanh \theta_i) = \delta_{ia} (\sigma_i - \tanh \theta_i). \quad (9)$$

For the coupling parameter J_{ab} with $a < b$, we account for symmetry ($J_{ab} = J_{ba}$), meaning J_{ab} enters the effective field θ_i when either $i = a$ or $i = b$:

$$\frac{\partial \theta_i}{\partial J_{ab}} = \delta_{ia} \sigma_b + \delta_{ib} \sigma_a. \quad (10)$$

This means:

$$\begin{aligned} \frac{\partial \ln P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i)}{\partial J_{ab}} &= \frac{\partial -\ln(1 + e^{-2\sigma_i \theta_i})}{\partial J_{ab}} \\ &= (\delta_{ia} \sigma_b + \delta_{ib} \sigma_a) \frac{2\sigma_i}{1 + e^{2\sigma_i \theta_i}} \\ &= (\delta_{ia} \sigma_b + \delta_{ib} \sigma_a) \sigma_i (1 - \sigma_i \tanh \theta_i) \\ &= (\delta_{ia} \sigma_b + \delta_{ib} \sigma_a) (\sigma_i - \tanh \theta_i). \end{aligned} \quad (11)$$

Summing over all sites i , the per-sample gradient with respect to J_{ab} becomes:

$$\sum_{i=1}^n \frac{\partial \ln P(\sigma_i | \boldsymbol{\sigma} \setminus \sigma_i)}{\partial J_{ab}} = \sigma_b(\sigma_a - \tanh \theta_a) + \sigma_a(\sigma_b - \tanh \theta_b) = 2\sigma_a\sigma_b - \sigma_b \tanh \theta_a - \sigma_a \tanh \theta_b. \quad (12)$$

The pseudo log-likelihood objective function over a dataset of m independent configurations $\{\boldsymbol{\sigma}^k\}_{k=1}^m$ is defined as:

$$\mathcal{L}(\mathbf{h}, \mathbf{J}) = \frac{1}{m} \sum_{k=1}^m \sum_{i=1}^n \ln P(\sigma_i^k | \boldsymbol{\sigma}^k \setminus \sigma_i^k). \quad (13)$$

Using Equation 9 and Equation 12, the full gradients for the PLE are:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_a} = \frac{1}{m} \sum_{k=1}^m (\sigma_a^k - \tanh \theta_a^k) = \langle \sigma_a \rangle - \langle \tanh \theta_a \rangle, \quad (14)$$

$$\begin{aligned} \frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial J_{ab}} &= \frac{1}{m} \sum_{k=1}^m (2\sigma_a^k \sigma_b^k - \sigma_b^k \tanh \theta_a^k - \sigma_a^k \tanh \theta_b^k) \\ &= 2\langle \sigma_a \sigma_b \rangle - \langle \sigma_b \tanh \theta_a \rangle - \langle \sigma_a \tanh \theta_b \rangle, \end{aligned} \quad (15)$$

where $\langle \cdot \rangle$ denotes the empirical sample average over the m configurations.

2 Potts Model

This section presents the mathematical derivation of the gradients for the pseudo likelihood estimation (PLE) of the generalized Potts model. Consider a system of n sites, where each site variable takes one of q states, denoted by $\boldsymbol{\sigma} \in \{1, 2, \dots, q\}^n$. The Hamiltonian of the system is defined as:

$$H(\boldsymbol{\sigma}) = - \sum_{i=1}^n h_i(\sigma_i) - \sum_{i=1}^n \sum_{j < i} J_{ij}(\sigma_i, \sigma_j), \quad (16)$$

which, according to the Boltzmann distribution, yields the joint probability:

$$p(\boldsymbol{\sigma}) = \frac{1}{Z} e^{-H(\boldsymbol{\sigma})} = \frac{1}{Z} \exp \left(\sum_{i=1}^n h_i(\sigma_i) + \sum_{i=1}^n \sum_{j < i} J_{ij}(\sigma_i, \sigma_j) \right), \quad (17)$$

where $Z = \sum_{\boldsymbol{\sigma}} e^{-H(\boldsymbol{\sigma})}$ is the partition function.

Assuming symmetry of the coupling interactions ($J_{ij}(r, s) = J_{ji}(s, r)$ for all states r, s and $J_{ii} = 0$), the Hamiltonian can be decomposed with respect to site i :

$$H(\boldsymbol{\sigma}) = H(\boldsymbol{\sigma} \setminus \sigma_i) + H(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) = H(\boldsymbol{\sigma} \setminus \sigma_i) - h_i(\sigma_i) - \sum_{j \neq i} J_{ij}(\sigma_i, \sigma_j). \quad (18)$$

The conditional probability of observing state σ_i at site i , given the state of all remaining sites $\boldsymbol{\sigma} \setminus \sigma_i$, is obtained by:

$$P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) = \frac{P(\sigma_i, \boldsymbol{\sigma} \setminus \sigma_i)}{\sum_{r=1}^q P(\sigma_i = r, \boldsymbol{\sigma} \setminus \sigma_i)}. \quad (19)$$

Substituting the Boltzmann distribution, the conditional distribution simplifies to:

$$P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) = \frac{e^{-H(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i)}}{\sum_{r=1}^q e^{-H(\sigma_i = r \mid \boldsymbol{\sigma} \setminus \sigma_i)}} = \frac{e^{-H(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i)}}{Z_i}, \quad (20)$$

where $Z_i = \sum_{r=1}^q e^{-H(\sigma_i = r \mid \boldsymbol{\sigma} \setminus \sigma_i)}$ denotes the site-specific partition function. The log-conditional probability for site i is therefore:

$$\ln P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) = -H(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i) - \ln Z_i. \quad (21)$$

To optimize the pseudo log-likelihood, we first compute the gradient of Equation 21 with respect to $h_a(k)$ at site a for state k . Differentiating the two terms separately:

$$\frac{\partial (-H(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i))}{\partial h_a(k)} = \delta_{ia} \delta_{\sigma_i, k}, \quad (22)$$

$$\frac{\partial (-\ln Z_i)}{\partial h_a(k)} = -\frac{1}{Z_i} \sum_{r=1}^q e^{-H(\sigma_i = r \mid \boldsymbol{\sigma} \setminus \sigma_i)} \delta_{ia} \delta_{r, k} = -\delta_{ia} P(\sigma_a = k \mid \boldsymbol{\sigma} \setminus \sigma_a). \quad (23)$$

Combining both terms yields the single-site derivative:

$$\frac{\partial \ln P(\sigma_i \mid \boldsymbol{\sigma} \setminus \sigma_i)}{\partial h_a(k)} = \delta_{ia} \left(\delta_{\sigma_a, k} - P(\sigma_a = k \mid \boldsymbol{\sigma} \setminus \sigma_a) \right). \quad (24)$$

Next, consider the derivative with respect to the coupling parameter $J_{ab}(k, l)$ between sites a and b for states k and l . Due to symmetry ($J_{ab}(k, l) = J_{ba}(l, k)$), this coupling enters the Hamiltonian when either $i = a$ or $i = b$:

$$\frac{\partial(-H(\sigma_i | \boldsymbol{\sigma} \setminus \sigma_i))}{\partial J_{ab}(k, l)} = \delta_{ia} \delta_{\sigma_a, k} \delta_{\sigma_b, l} + \delta_{ib} \delta_{\sigma_b, l} \delta_{\sigma_a, k}, \quad (25)$$

$$\begin{aligned} \frac{\partial(-\ln Z_i)}{\partial J_{ab}(k, l)} &= -\frac{1}{Z_i} \sum_{r=1}^q e^{-H(\sigma_i=r | \boldsymbol{\sigma} \setminus \sigma_i)} (\delta_{ia} \delta_{r, k} \delta_{\sigma_b, l} + \delta_{ib} \delta_{r, l} \delta_{\sigma_a, k}) \\ &= -\delta_{ia} \delta_{\sigma_b, l} P(\sigma_a = k | \boldsymbol{\sigma} \setminus \sigma_a) - \delta_{ib} \delta_{\sigma_a, k} P(\sigma_b = l | \boldsymbol{\sigma} \setminus \sigma_b). \end{aligned} \quad (26)$$

Combining these terms gives the per-site gradient for the coupling parameter:

$$\begin{aligned} \frac{\partial \ln P(\sigma_i | \boldsymbol{\sigma} \setminus \sigma_i)}{\partial J_{ab}(k, l)} &= \delta_{ia} \delta_{\sigma_b, l} (\delta_{\sigma_a, k} - P(\sigma_a = k | \boldsymbol{\sigma} \setminus \sigma_a)) \\ &\quad + \delta_{ib} \delta_{\sigma_a, k} (\delta_{\sigma_b, l} - P(\sigma_b = l | \boldsymbol{\sigma} \setminus \sigma_b)). \end{aligned} \quad (27)$$

Summing over all sites $i = 1, \dots, n$ gets the contributions from both conditional probabilities where $J_{ab}(k, l)$ is active:

$$\sum_{i=1}^n \frac{\partial \ln P(\sigma_i | \boldsymbol{\sigma} \setminus \sigma_i)}{\partial J_{ab}(k, l)} = 2\delta_{\sigma_a, k} \delta_{\sigma_b, l} - \delta_{\sigma_b, l} P(\sigma_a = k | \boldsymbol{\sigma} \setminus \sigma_a) - \delta_{\sigma_a, k} P(\sigma_b = l | \boldsymbol{\sigma} \setminus \sigma_b). \quad (28)$$

Given a dataset of M independent configurations $\{\boldsymbol{\sigma}^m\}_{m=1}^M$, the total pseudo log-likelihood is:

$$\mathcal{L}(\mathbf{h}, \mathbf{J}) = \frac{1}{M} \sum_{m=1}^M \sum_{i=1}^n \ln P(\sigma_i^m | \boldsymbol{\sigma}^m \setminus \sigma_i^m). \quad (29)$$

Using Equation 24 and Equation 28, the full gradients for the Potts model pseudo likelihood estimation are:

$$\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial h_a(k)} = \frac{1}{M} \sum_{m=1}^M (\delta_{\sigma_a^m, k} - P(\sigma_a = k | \boldsymbol{\sigma}^m \setminus \sigma_a^m)) = \langle \delta_{\sigma_a, k} \rangle - \langle P(\sigma_a = k | \boldsymbol{\sigma} \setminus \sigma_a) \rangle, \quad (30)$$

$$\begin{aligned}
\frac{\partial \mathcal{L}(\mathbf{h}, \mathbf{J})}{\partial J_{ab}(k, l)} &= \frac{1}{M} \sum_{m=1}^M \left(2\delta_{\sigma_a^m, k} \delta_{\sigma_b^m, l} - \delta_{\sigma_b^m, l} P(\sigma_a = k \mid \boldsymbol{\sigma}^m \setminus \sigma_a^m) - \delta_{\sigma_a^m, k} P(\sigma_b = l \mid \boldsymbol{\sigma}^m \setminus \sigma_b^m) \right) \\
&= 2\langle \delta_{\sigma_a, k} \delta_{\sigma_b, l} \rangle - \langle \delta_{\sigma_b, l} P(\sigma_a = k \mid \boldsymbol{\sigma} \setminus \sigma_a) \rangle - \langle \delta_{\sigma_a, k} P(\sigma_b = l \mid \boldsymbol{\sigma} \setminus \sigma_b) \rangle,
\end{aligned} \tag{31}$$

where $\langle \cdot \rangle$ denotes the empirical average across the M observed sequences.

3 Software Implementation

For computational efficiency and parallelization in **spininfer**, each site a independently accumulates its directed conditional gradient:

$$G_{ab} \equiv \frac{\partial \mathcal{L}_a}{\partial J_{ab}} = \langle \sigma_a \sigma_b \rangle - \langle \sigma_b \tanh \theta_a \rangle, \tag{32}$$

and for the Potts model:

$$G_{ab}(k, l) \equiv \frac{\partial \mathcal{L}_a}{\partial J_{ab}(k, l)} = \langle \delta_{\sigma_a, k} \delta_{\sigma_b, l} \rangle - \langle \delta_{\sigma_b, l} P(\sigma_a = k \mid \boldsymbol{\sigma} \setminus \sigma_a) \rangle. \tag{33}$$

To preserve coupling symmetry under gradient updates, the full gradient is subsequently symmetrized according to:

$$G_{ab}^{\text{sym}} = \frac{G_{ab} + G_{ba}}{2}, \quad G_{ab}^{\text{sym}}(k, l) = \frac{G_{ab}(k, l) + G_{ba}(l, k)}{2}. \tag{34}$$

Note that this is identical to the gradient of the total pseudo likelihood with a constant prefactor of $1/2$.